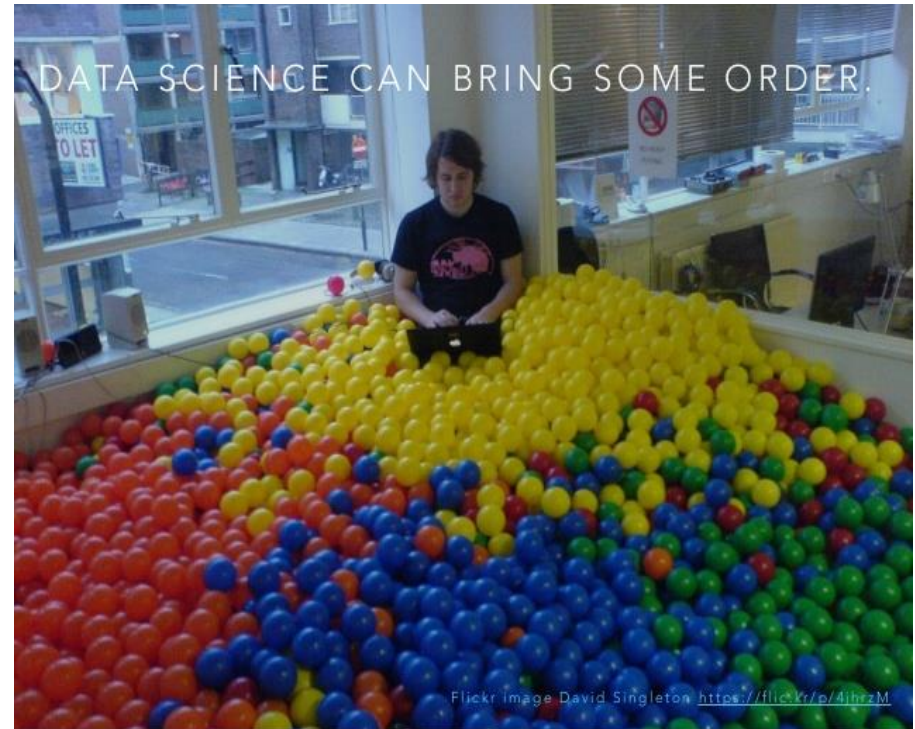




Introduction to Data Science:

Data Science & Big Data

What is Data Science?



What is Data Science?

- Data Science, in general terms, is the extraction of knowledge from data.
- Then use that knowledge to do the following:
 - Understand the past/present
 - Make decisions
 - Predict the future
 - Create new industries/products
- A practitioner of data science is called a data scientist.

History of Data Science

1960

- The term "data science" (originally used interchangeably with "datalogy") has existed for over thirty years and was used initially as a substitute for computer science by Peter Naur in 1960.

2002

- The International Council for Science: Committee on Data for Science and Technology started the Data Science Journal

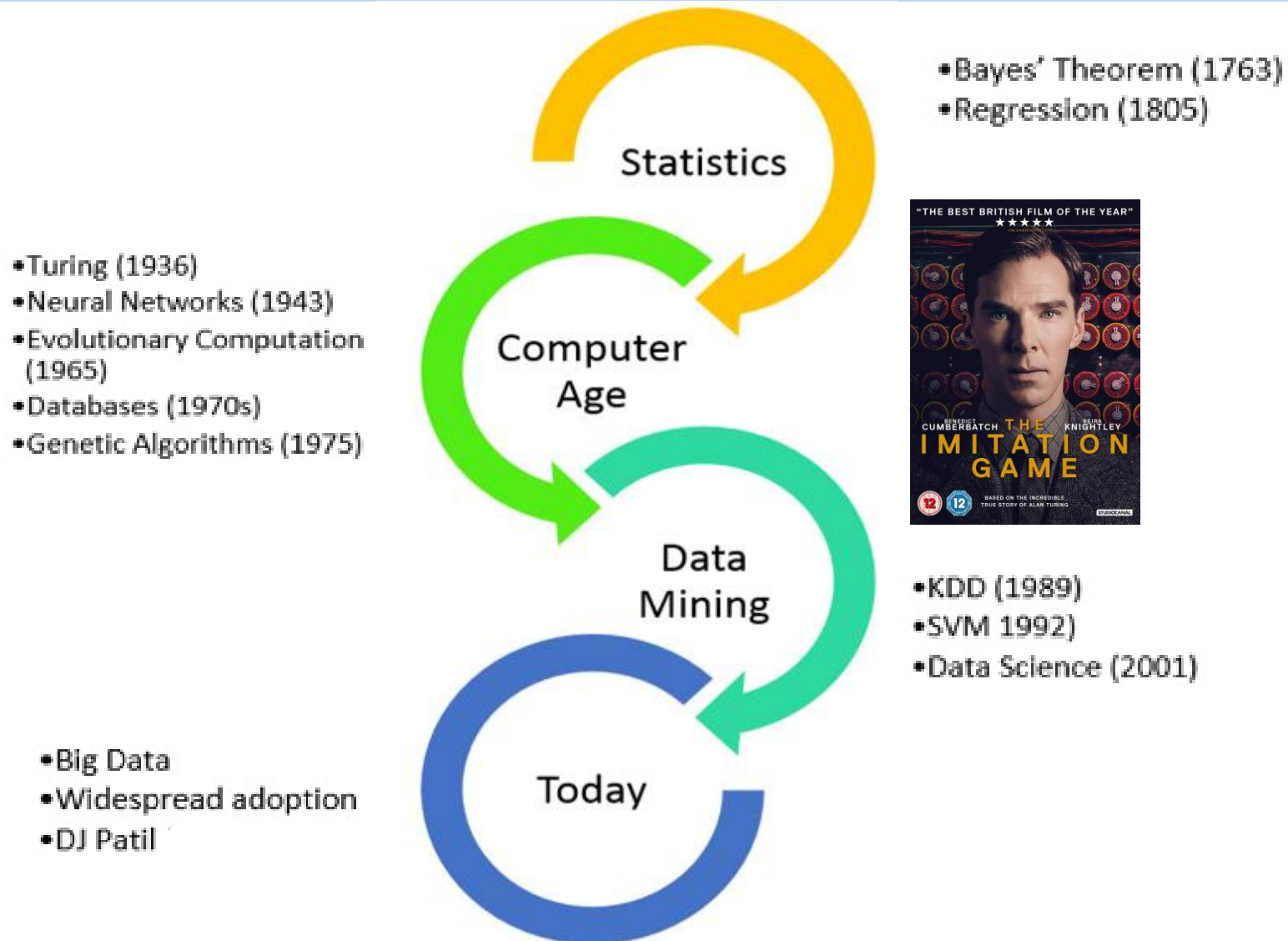
2004

- Usama Fayyad became the first CDO (Chief Data Officer) at Yahoo.

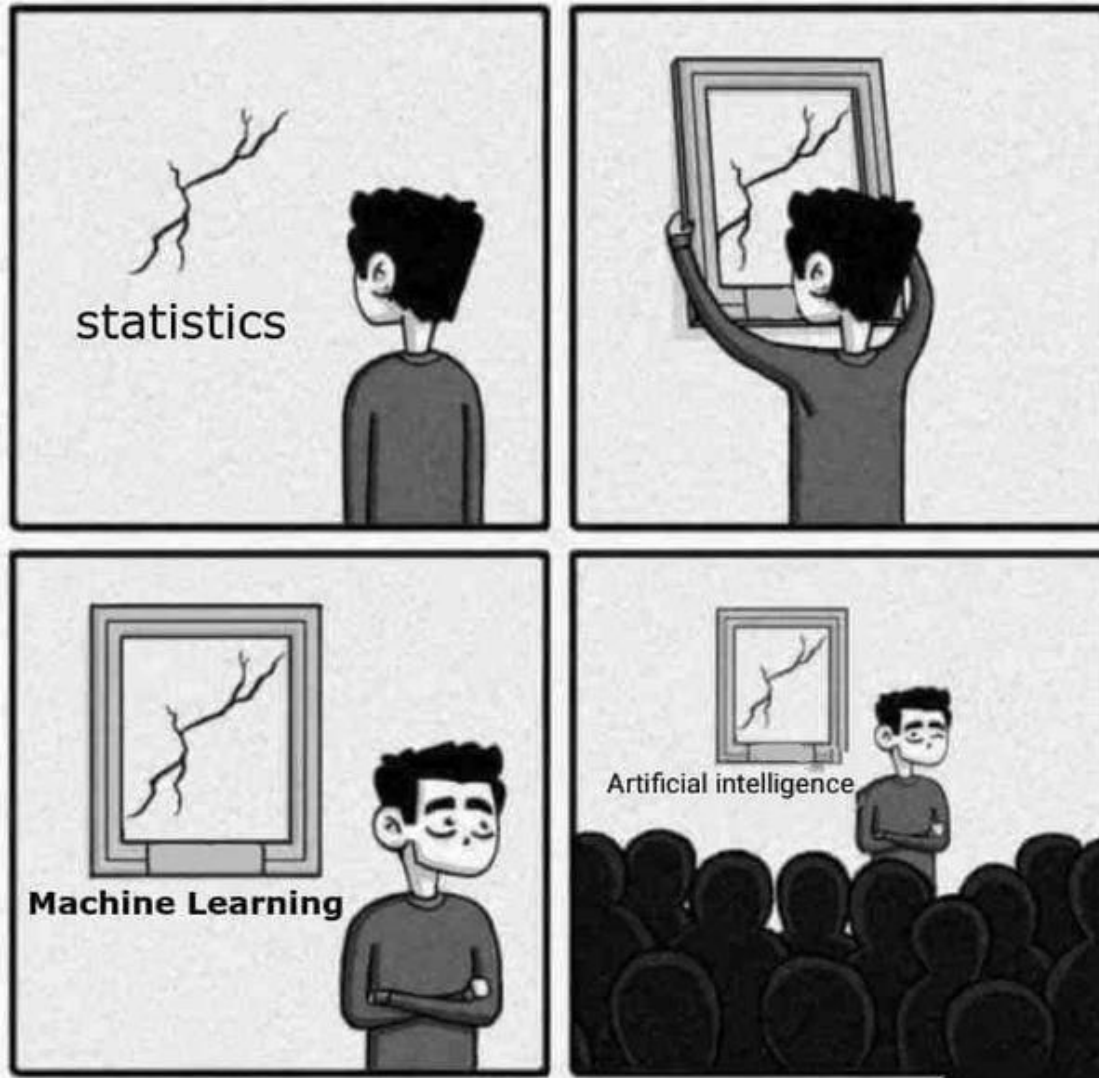
2008

- DJ Patil and Jeff Hammerbacher coined the term "data scientist" to define their jobs at LinkedIn and Facebook, respectively

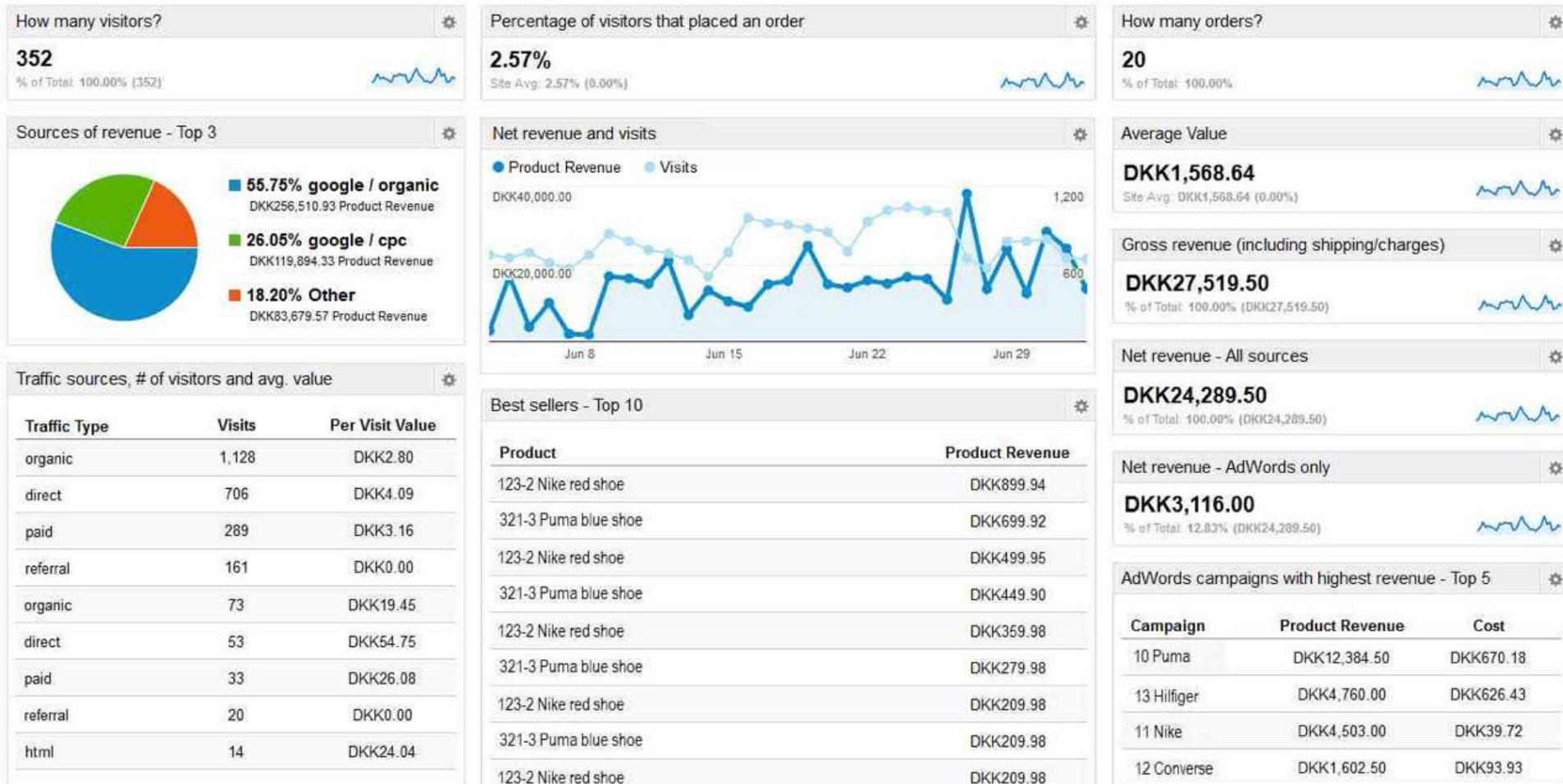
Evolution of Data Science



Evolution of Data Science



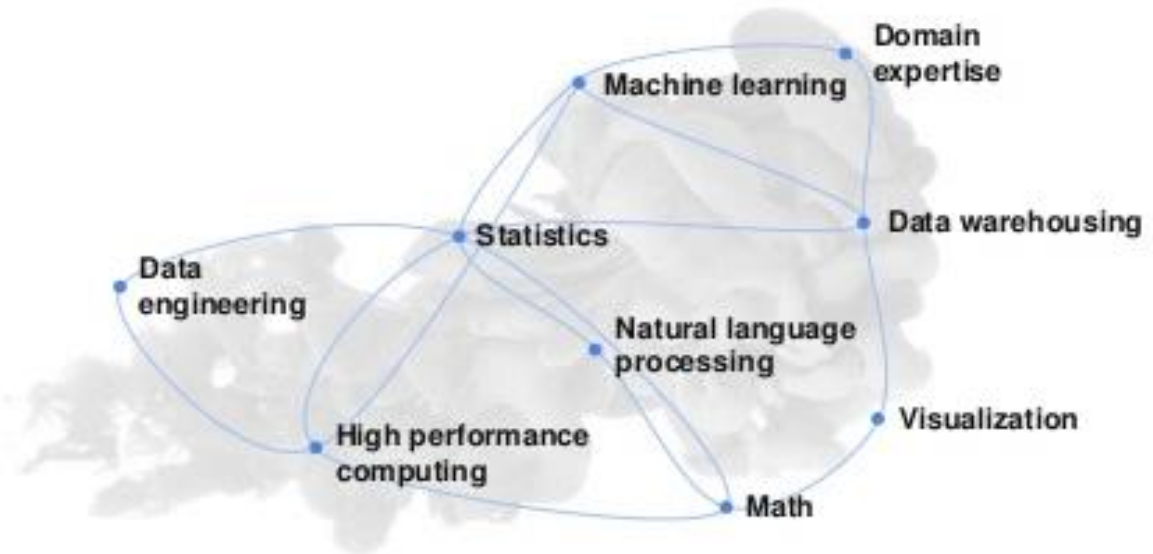
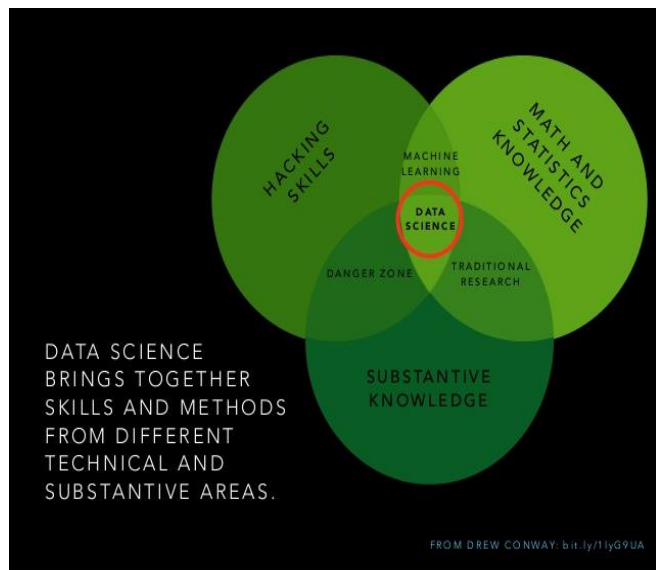
Is this Data Science?



Data Scientist as a Unicorn

Requires multi-disciplinary skillsets to be able to derive actionable insights from data to solve problems.

Data Science Skills



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Using Data Science to solve problems in

Are there any sectors that are still untouched by AI and DS?



Made possible with Data



Ethical concerns



*Just because you can....
doesn't mean you need to...*

Types of Data

UNSTRUCTURED DATA



Images



Text



Videos



Audio

Harder to process; generally need to process and analyse further.

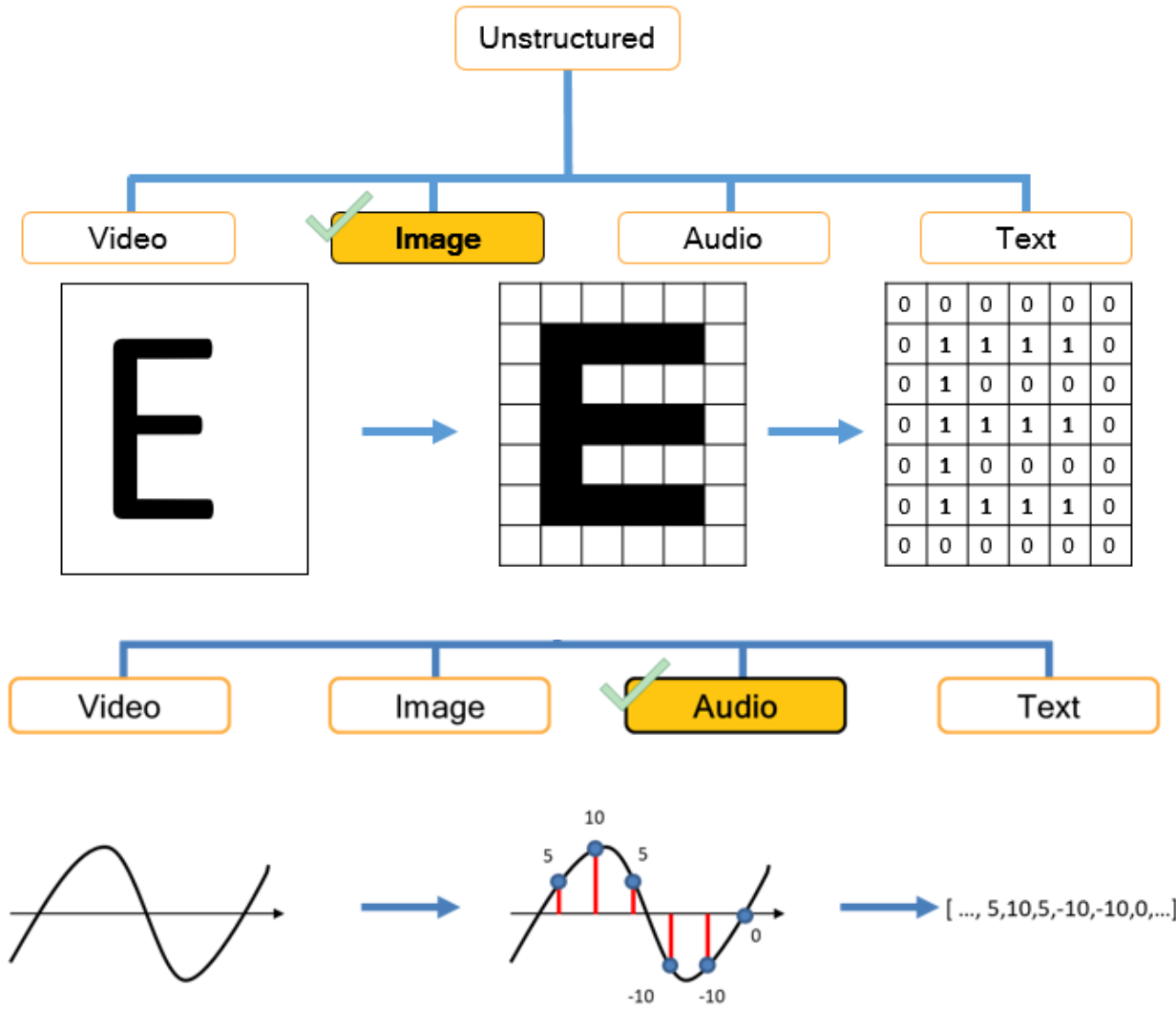
STRUCTURED DATA



Tabular Data

Some consistency so is easier to process and analyse.

Pervasive use of numerical packages, e.g. numpy



What is Big Data?



What is Big Data?

“Large pools of data that can be captured, communicated, aggregated, stored and analyzed” *McKinsey Global Institute*

“A collection of data from traditional and digital sources inside and outside your company that represents a source for ongoing discovery and analysis” *What is Big Data, Lisa Arthur, Forbes*

Big Data is subjected to 4Vs

4 V's of Big Data

Volume

- Data quantity

Velocity

- Data Speed

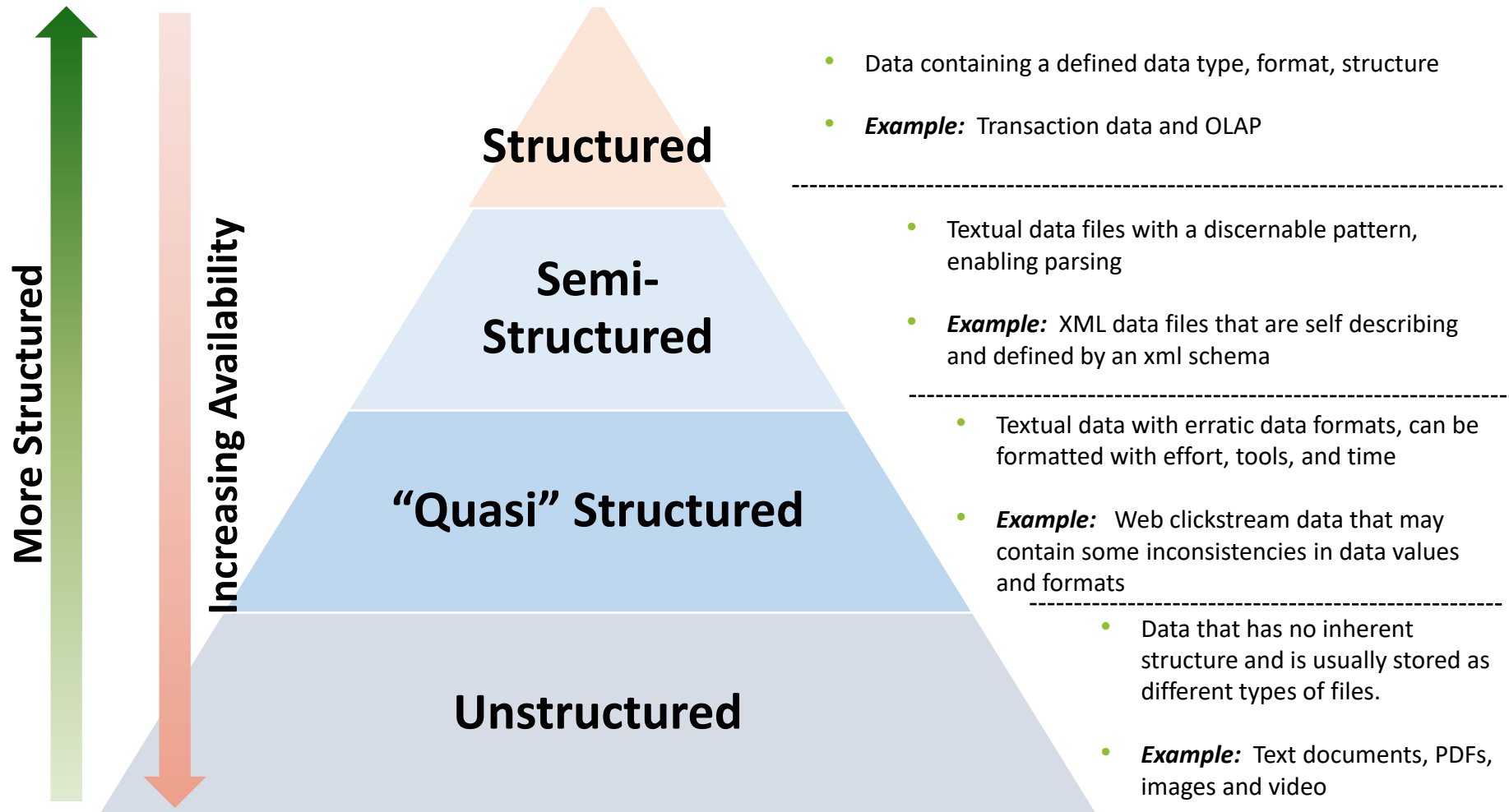
Variety

- Data Types

Veracity

- Messiness

Different structures of Big Data



Different structures of Big Data

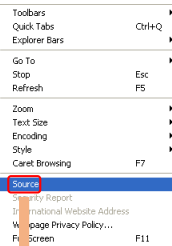
Structured Data

SUMMER FOOD SERVICE PROGRAM 1)				
(Data as of August 01, 2011)				
Fiscal Year	Number of Sites	Peak (July) Participation	Meals Served	Total Federal Expenditures 2)
	-----Thousands-----		--Mil--	---Million \$---
1969	1.2	99	2.2	0.3
1970	1.9	227	8.2	1.8
1971	3.2	569	29.0	8.2
1972	6.5	1,080	73.5	21.9
1973	11.2	1,437	65.4	26.6
1974	10.6	1,403	63.6	33.6
1975	12.0	1,785	84.3	50.3
1976	16.0	2,453	104.8	73.4
TQ 3)	22.4	3,455	198.0	88.9
1977	23.7	2,791	170.4	114.4
1978	22.4	2,333	120.3	100.3
1979	23.0	2,126	121.8	108.6
1980	21.6	1,922	108.2	110.1

Semi-Structured Data

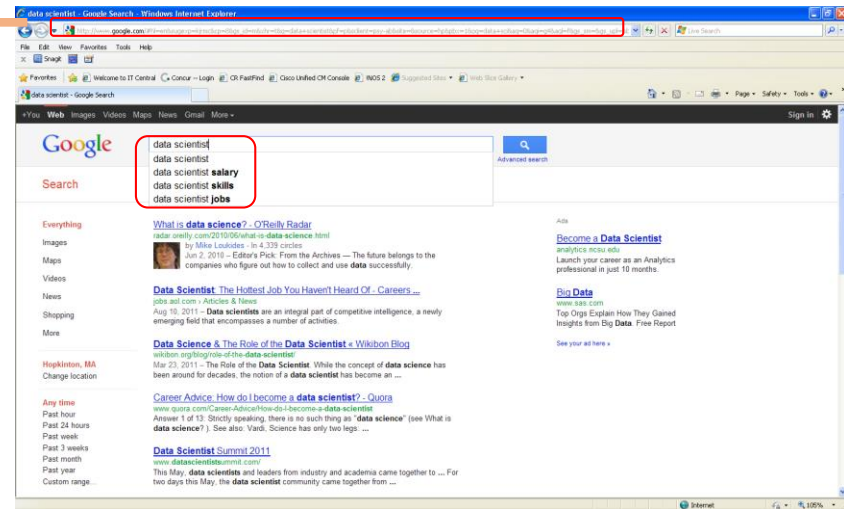


View → Source



```
1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-trans:
2 <html xmlns="http://www.w3.org/1999/xhtml">
3
4
5
6
7 <head>
8   <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
9   <META name="y_key" content="859b4020e1c9aec">
10  <link rel="canonical" href="http://www.emc.com/index.htm" />
11  <META NAME="verify-v1" CONTENT="vi2t9V084eV0jFdIpeVvIrfR32g4gtwFE0i2UvTMfSU
12  <title>EMC - Data Recovery, Cloud Computing, and Storage Hardware</title>
13  <META NAME="description" CONTENT="EMC is a leading provider of storage hardware solutions th
14  data recovery and improve cloud computing." />
15  <META NAME="keywords" CONTENT="emc, network storage, data recovery, information manage
16  software, nas storage, information protection, information management" />
17  <!-- Start stylesheet includes -->
18  <link rel="stylesheet" href="/_admin/css/styles.css" />
19  <link rel="stylesheet" href="/_admin/css/styles_nav.css" />
20  <!--[if IE]>
```

Quasi-Structured Data



http://www.google.com/#hl=en&sugexp=jkrmc&cp=8&gs_id=2m&xhr=t&q=data+scientist&pq=big+data&pf=p&scient=psyb&source=hp&pbx=1&oq=data+sci&aq=0&aql=f&gs_sm=&gs_upl=&bav-on.2,or,r_gc,r_pw,,cf.osb&fp=d566e0fbd09c8604&biw=1382&bih=651

Unstructured Data

The Red Wheelbarrow, by
William Carlos Williams

so much depends
upon

a red wheel
barrow

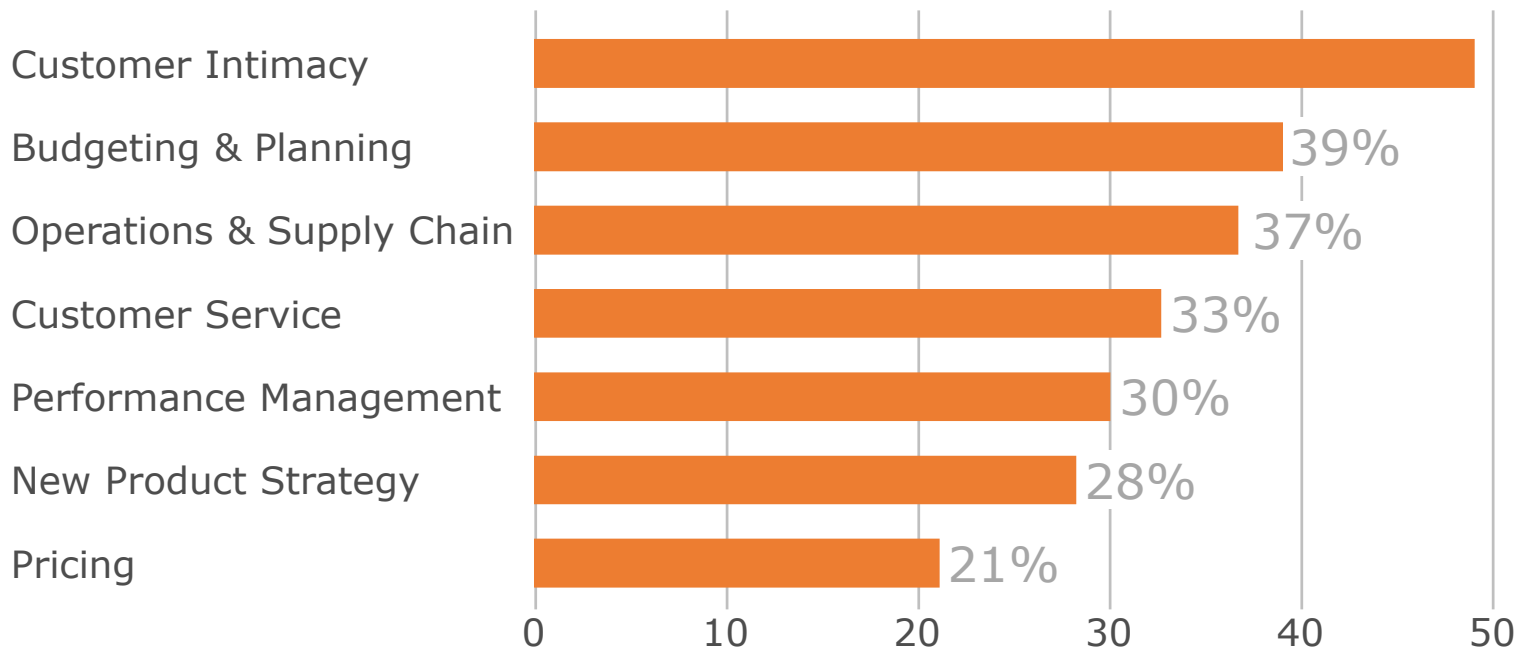
glazed with rain
water

beside the white
chickens.



Who consumes Big Data

- Functional Areas Where Companies Are Using Big Data



McKinsey Global Survey of 1,469 C-level executive respondents at a range of industries and company sizes, "Minding Your Digital Business," 2012.

**WE 
DATA
SCIENCE**